

Research Paper

One-Step Analysis for *Listeria monocytogenes* Growth in Ready-to-Eat Braised Beef at Dynamic and Static Conditions

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ABSTRACT

This study aimed to estimate the growth parameters of *Listeria monocytogenes* growth in ready-to-eat (RTE) braised beef by one-step dynamic and static kinetic analysis. The Baranyi model and cardinal parameters model were integrated into a dynamic and static model to estimate the kinetic parameters under one dynamic condition (−20 to 40.0°C) and eight static conditions (4, 8, 15, 20, 30, 35, 37, and 40°C). Based on the dynamic and static methods, the respective dynamic and static results for estimated growth boundaries of *L. monocytogenes* in RTE braised beef were from −2.5 and −2.7°C to 40.5 and 40.7°C with optimal specific growth rates of 1.078 and 0.913 per h at temperatures of 35.7 and 35.0°C. Temperature effects on the specific growth rate and lag period were developed and used to simulate the change of the physiological state of inocula during the bacterial growth. Subsequently, three additional dynamic temperature profiles were implemented for external validation. The root mean square error of the model developed by dynamic regression (0.19 log CFU/g) is slightly better than that of the model developed by static regression (0.23 log CFU/g). Comparing the validation results, one-step dynamic analysis might be a preferable method for prediction, especially when the growth approaches the stationary phase. Generally, both one-step dynamic and static analyses could be used to accurately predict *L. monocytogenes* growth in RTE braised beef under fluctuating temperatures.

HIGHLIGHTS

- Growth of *L. monocytogenes* in RTE beef under dynamic and static conditions was studied.
- *L. monocytogenes* growth parameters were obtained by different one-step analyses.
- Secondary models were developed and used to simulate the physiological state.
- External validations were applied in additional dynamic conditions.

Key words: Fluctuating temperature; Meat products; Numerical method; Physiological state; Predictive microbiology

Listeria monocytogenes is a pathogenic bacterium and ubiquitously distributed in the food supply chain (6, 21). Ready-to-eat (RTE) meat products are usually perceived as one of the main sources of *L. monocytogenes* infections (10, 19). A severe outbreak of listeriosis, leading to more than 200 deaths in South Africa, was confirmed to be connected with consuming a *L. monocytogenes*-contaminated RTE meat product (42). RTE braised beef is one of the common RTE meat products in the People's Republic of China that can potentially support *L. monocytogenes* growth. It is necessary to model *L. monocytogenes* growth dynamics based on the kinetic analysis methodology, thereby predicting its contamination level in RTE braised beef during the supply process.

A two-step approach has been widely used to determine the growth and inactivation parameters based on multiple

isothermal tests (14, 20, 41). During this process, the primary model is implemented to describe the change of the bacterial concentration and obtain the kinetic parameters, namely, lag time and specific growth (inactivation) rate, under a certain condition. Then, the secondary model is used to describe the effect of intrinsic and extrinsic factors (e.g., temperature, pH, water activity, or nitrite) on the kinetic parameters (14, 41). However, the two-step methodology is not able to consider the overall error of the estimated parameters derived from the primary and secondary model, which may cause a deviation when applying the estimates under some dynamic and extreme environmental conditions (20, 24). In contrast, a one-step approach constructs the primary and secondary models as one integrated model and thus obtains all growth and inactivation parameters in one analysis. The one-step approach can globally minimize the error between predictions and observations, which may improve the accuracy of regression (14, 24, 36). The one-step kinetic analysis has been applied not only under isothermal (static) conditions

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but also under nonisothermal (dynamic) conditions by introducing a numerical method (13, 14). Cattani et al. (8) and Huang (12) pointed out that the one-step dynamic method could provide more accurate estimates with fewer tests than the static method for the microbial inactivation kinetic analysis. Nevertheless, few studies have compared the kinetic growth analysis between the one-step dynamic method and the one-step static method.

Therefore, the objective of this study was to (i) obtain growth parameters by the one-step approach under dynamic and static conditions, (ii) compare the estimation derived from the one-step dynamic and static methods, (iii) simulate the temperature effect on the kinetic parameter, and (iv) validate the fitted parameters under additional dynamic conditions for *L. monocytogenes* in RTE braised beef.

MATERIALS AND METHODS

Bacterial strains and inoculum preparation. *L. monocytogenes* strains of four serotypes, 1/2a, 1/2b, 1/2c, and 4b (Chinese meat products isolated), were first streaked from frozen stocks (stored at -80°C) onto Trypticase soy agar supplemented with 0.6% yeast extract (TSAYE; Hope Bio-Technology Co., Qingdao, China) and stored separately at 4°C for working stocks. The strains were activated individually by transferring a loopful from TSAYE into 100 mL of Trypticase soy broth supplemented with 0.6% yeast extract (Hope Bio-Technology) and incubated at 37°C for 14 h to reach the late stationary phase. Each 5-mL enriched culture was centrifuged at $4,300 \times g$ and 4°C for 10 min and washed twice by 5 mL of sterile 0.9% (w/v) saline solution to establish a 20-mL cocktail suspension.

Sample preparation, inoculation, and incubation. Pingyao braised beef, a geographical RTE beef product of Shanxi, China, was used to build the experimental matrix. The prepackaged beef sample (25 g per portion) containing ca. 1% NaCl was purchased from the local supermarket in Shanghai, China. The pH and water activity of RTE braise beef were approximately 6.2 and 0.983, respectively. The samples were first divided into sterile stomacher bags and subjected to a heat shock treatment at 80°C for 20 min to eliminate vegetative cells. Then, a volume of 0.2 mL of diluted cocktail suspension of *L. monocytogenes* was spread on each

portion to obtain an initial concentration from 3 to 4 log CFU/g. Uninoculated samples were used as negative controls.

All samples were aseptically air packaged and transferred into the programmed alternating temperature chamber (BPHJ-060A; Yiheng Scientific Instruments Co., Shanghai, China). For model development, the inoculated beef samples were incubated under one dynamic and eight static temperature conditions. The temperature range of the fluctuating temperature condition (FTC), named FTC1 (-20 to $40 \pm 1^{\circ}\text{C}$, during 150 h), was determined based on the sensitivity analysis approach (data not shown) for the dynamic experiment design mentioned in previous research (5, 38, 39). The cooling and heating rate (from 3 to $20^{\circ}\text{C}/\text{h}$) was set by considering the real food chain process (26). In addition, exposing the bacteria to sublethal low temperatures (-20°C for *L. monocytogenes* in this study) may help accurately estimate theoretical minimum temperature for growth (15). The static experiments were conducted at 4, 8, 15, 20, 30, 35, 37, and $40 \pm 1^{\circ}\text{C}$. For model validation, three additional dynamic temperature profiles were accomplished in the chamber as FTC2 (5.4 to $40 \pm 1^{\circ}\text{C}$, during 52 h) and FTC3 (4 to $30 \pm 1^{\circ}\text{C}$, during 150 h) or in the refrigerator or at room temperature as FTC4 (8 to $27.7 \pm 1^{\circ}\text{C}$, during 33 h). Trials were repeated twice under the conditions of FTC2 and FTC3. The temperature was monitored and recorded every 6 min during the incubation period by a mini-temperature logger (RC-5+; Elitech Co., Jiangsu, China).

***L. monocytogenes* enumeration.** At appropriate intervals, three (duplicate) samples were retrieved for bacterial counting. During the enumeration, a volume of 225 mL of 0.9% saline was added to each incubated sample bag and homogenized in a stomacher (SCIENTZ-09; Scientz Biotechnology Co., Ningbo, China) for 2 min at 6.5 times per s. Then, each sample was 10-fold diluted with 0.9% saline in the 96-well plate and surface plated on a Palfam plate (Hope Bio-Technology) using the drop plate method (9). All plates were incubated at $35 \pm 1^{\circ}\text{C}$ for 24 h before the enumeration.

Mathematical modeling. The complete Baranyi model (3) and cardinal parameters model (34) were constructed as the integrated model for one-step analysis in this study. A set of ordinary differential equations was used to describe the dynamic growth of *L. monocytogenes* in RTE braised beef (equation 1):

$$\begin{cases} \frac{dY}{dt} = \mu_{\max} \frac{q}{1+q} [1 - \exp(Y - Y_{\max})] \\ \frac{dq}{dt} = \mu_{\max} q \\ \mu_{\max} = \begin{cases} 0, & T \leq T_{\min} \\ \frac{\mu_{\text{opt}}(T - T_{\max})(T - T_{\min})^2}{[(T_{\text{opt}} - T_{\min})(T - T_{\text{opt}}) - (T_{\text{opt}} - T_{\max})(T_{\text{opt}} + T_{\min} - 2T)](T_{\text{opt}} - T_{\min})}, & T_{\min} < T \leq T_{\max} \\ 0, & T > T_{\max} \end{cases} \\ Y(0) = Y_0 \\ q(0) = q_0 \end{cases} \quad (1)$$

where Y is the natural logarithm of bacterial concentration at time t (ln CFU/g), $Y_{\max}$ is the natural logarithm of the maximum population density (MPD) of the bacteria (ln CFU/g), q is the physiological state of the inoculum, and T is the real-time temperature ($^{\circ}\text{C}$). The variables $\mu_{\max}$ and μ_{opt} are the maximum and optimal specific growth rates (per hour). The variables $T_{\min}$, $T_{\max}$, and T_{opt} represent the minimum, maximum, and optimal temperature for microbial growth ($^{\circ}\text{C}$). The variables Y_0 and q_0 are the initial values of Y and q , respectively. Because microbial

inactivation is not considered, the value of $\mu_{\max}$ is assumed to be zero beyond the growth temperature range.

The ordinary differential equation in the dynamic model cannot be solved with an analytical solution (13). The traditional fourth-order Runge-Kutta method is an improved Euler method that has been widely applied to numerically solve microbial ordinary differential equation problems in previous studies (16, 22, 25, 31). Thus, the fourth-order Runge-Kutta method described by equation 2 was implemented to analyze and simulate the dynamic growth:

$$\begin{cases} Y_{n+1} = Y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4) \\ k_1 = f(t_n, Y_n) \\ k_2 = f(t_n + \frac{h}{2}, Y_n + \frac{h}{2}k_1) \\ k_3 = f(t_n + \frac{h}{2}, Y_n + \frac{h}{2}k_2) \\ k_4 = f(t_n + h, Y_n + hk_3) \end{cases} \quad (2)$$

where Y_n is the approximation at t_n (ln CFU/g); h is the time step, which is set to 0.1 h in this study; $f(*)$ is the function of the ordinary differential equation, which is $dY(t)/dt$ or $dq(t)/dt$ in equation 1; and k_1 , k_2 , k_3 , and k_4 represent different increments calculated by $f(*)$ at each step.

For the one-step analysis under static conditions, the global static model can be written as follows:

$$\begin{cases} Y = Y_0 + A(t) - \ln \left[1 + \frac{\exp[A(t)] - 1}{\exp(Y_{\max} - Y_0)} \right] \\ A(t) = \mu_{\max} t + \ln[\exp(-\mu_{\max} t) + \exp(-\mu_{\max} \lambda) - \exp(-\mu_{\max} t - \mu_{\max} \lambda)] \\ \mu_{\max} = \begin{cases} 0, & T \leq T_{\min} \\ \frac{\mu_{\text{opt}}(T - T_{\min})(T - T_{\max})^2}{[(T_{\text{opt}} - T_{\min})(T - T_{\text{opt}}) - (T_{\text{opt}} - T_{\max})(T_{\text{opt}} + T_{\min} - 2T)](T_{\text{opt}} - T_{\min})}, & T_{\min} < T \leq T_{\max} \\ 0, & T > T_{\max} \end{cases} \\ \lambda = \frac{\ln \left(1 + \frac{1}{q_0} \right)}{\mu_{\max}} \end{cases} \quad (3)$$

where λ is the lag time under the static condition (h). The equation of λ indicates that its value is decided by not only the current environmental condition (via $\mu_{\max}$) but also the historical condition (via q_0) (3, 17). Because a consistent inoculation process was implemented before static incubation, the values of Y_0 and q_0 were assumed to be equal for each experimental inoculum in the one-step static analysis.

Moreover, Baranyi and Roberts (3) defined q as the ratio of the so-called critical substance concentration and the Michaelis-Menten constant. Based on this hypothesis, the change of q can be simulated by the differential equation of q with the value of q_0 and the temperature profile. Because the upper limit of q tends toward positive infinity, it is reasonable to transfer q into a limited function of α by equation 4 for simulation (3):

$$\alpha = \frac{q}{1 + q} \quad (4)$$

The preceding one-step modeling and numerical analysis were performed in R statistical software (version 3.4.3 for Mac OS X; <http://www.r-project.org>). A nonlinear least-squares function of *nls* in the stats package was used to determine parameters by minimizing the errors between fitted values (numerical solutions) and observations (16, 20). The estimation groups of dynamic parameters (DPs) and static parameters (SPs), including q_0 , Y_0 , $Y_{\max}$, $T_{\min}$, $T_{\max}$, T_{opt} , and μ_{opt} , were obtained by the one-step dynamic and static methods, respectively. Once both parameter groups were estimated, the secondary model of $\mu_{\max}$ and λ could be established, and the change of α could be simulated.

Indices for model evaluation and validation. The goodness of fit of the integrated models was evaluated by the corrected Akaike information criterion (AICc) (37) and root mean square error (RMSE) (29) as follows:

$$\text{AICc} = n \ln \left(\frac{\sum_{i=1}^n (\hat{Y}_i - Y_i)^2}{n} \right) + 2(p + 1) + \frac{2(p + 1)(p + 2)}{n - p - 2} \quad (5)$$

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{n - p}} \quad (6)$$

where n is the number of sample data, p is the number of parameters, $\hat{Y}_i$ (ln CFU/g) is the fitted bacterial concentration, and Y_i (ln CFU/g) is the observed bacterial concentration.

In the validation step, both the RMSE and the bias factor (33) were applied to compare the prediction accuracy of *L. monocytogenes* growth in RTE braised beef under the additional dynamic conditions. A bias factor (BF) value greater than 1 indicates a fail-dangerous model (33):

$$\text{BF} = 10^{\frac{1}{n} \sum_{i=1}^n \log(Y_i / \hat{Y}_i)} \quad (7)$$

Statistical analysis. The growth data were analyzed by using analysis of variance, which was performed in R by the *aov* function in the stats package. Statistical significance was analyzed by *t* test at a significance level of 0.05.

RESULTS AND DISCUSSION

Parameter determination and model evaluation. A total of 72 and 213 data of the inoculated samples were collected under FTC1 and eight static conditions for model development, respectively. No naturally occurring *L. monocytogenes* were found in the negative controls. The significant analysis of the initial enumeration value had no statistical difference between static experiments ($P = 0.158$).

The measurements of the time-temperature profile and bacterial concentration in Figures 1 and 2 reveal that *L. monocytogenes* can grow in RTE braised beef in a wide temperature range. As presented in Figure 1, *L. monocytogenes* shows delayed and slow growth at the initial refrigeration temperature (0 to 8°C) and proliferates rapidly under the favorable temperature. The subzero temperature period of FTC1 (as low as −20°C) had no significant effect on the bacterial concentration, which implied that *L. monocytogenes* might survive for a long time under low temperatures in RTE braised beef. Similar phenomena were described by Ben Slama et al. (4) in a cheese matrix. Figure 2 illustrates that the MPD of *L. monocytogenes* can reach up to 8.20 log CFU/g under most of static conditions in RTE braised beef but not exceed 7.60 log CFU/g at the suboptimal temperature of 4 and 8°C. It was indicated that

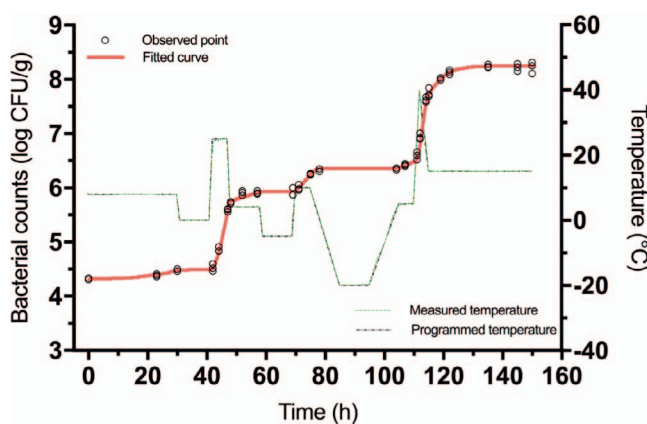


FIGURE 1. One-step fitting of the dynamic model under FTC1. Programmed time (hours) = 0, 30, 31, 41, 42, 47, 48, 58, 59, 69, 70, 75, 85, 95, 105, 110, 112, 115, and 150; programmed temperature (°C) = 8, 8, 0, 0, 25, 25, 4, 4, -5, -5, 10, 10, -20, 5, 5, 40, 15, and 15.

the isothermal low-temperature condition probably restricts the carrying capacity of meat products for the *L. monocytogenes* microorganism (7, 30).

The estimations of DP and SP are listed in Tables 1 and 2, respectively. Because of a similar inoculation preparation process, the initial conditions of q_0 (0.021 and 0.031) and Y_0 (9.94 and 9.85 ln CFU/g or 4.31 and 4.28 log CFU/g) were close in both estimations. However, the estimated q_0 only represented the initial status of the experimental inoculum, which might differ from that of the natural microorganism in contaminated samples. The theoretical $T_{\min}$ of *L. monocytogenes* in RTE braised beef was estimated to be -2.5 and -2.7°C, lower than that of previously reported values (-1.5 to -0.4°C) in most other foods (18, 23). One possible explanation for this result is that cooked beef is carnitine-rich food, which may help the *sigB* gene enhance *L. monocytogenes* survival and growth at chill temperatures (1, 27, 32, 35). The estimated value of μ_{opt} in DP (1.078/h) showed a larger value than in SP (0.913/h) but with a similar theoretical value of T_{opt} (35.7 and 35.0°C, respectively). Here, we determined μ_{max} at 35°C by fitting the static Baranyi model to the corresponding observation in R. The isothermal estimation of μ_{max} at 35°C was 1.062/h, which was closer to the estimated value of μ_{opt} in DP.

Both methods showed a low standard error on the Baranyi model parameter (Y_0 , Y_{max} , and q_0). However, the standard error of the secondary parameters ($T_{\min}$, T_{max} , T_{opt} , and μ_{opt}), were lower in the one-step static regression than in the one-step dynamic regression. Part of the reason was that more observations were collected during the static experiment than the dynamic experiment under different temperatures. In particular, the measurements of bacterial count near the optimum and maximum temperature were insufficient in FTC1, which might cause inaccurate estimation of T_{max} , T_{opt} , and μ_{opt} . In addition, the relative errors of q_0 (23.44% = 0.0049 and 0.0209) and $T_{\min}$ (25.98% = 0.66 and 2.54) were high, which implied both parameters were more difficult to estimate than other

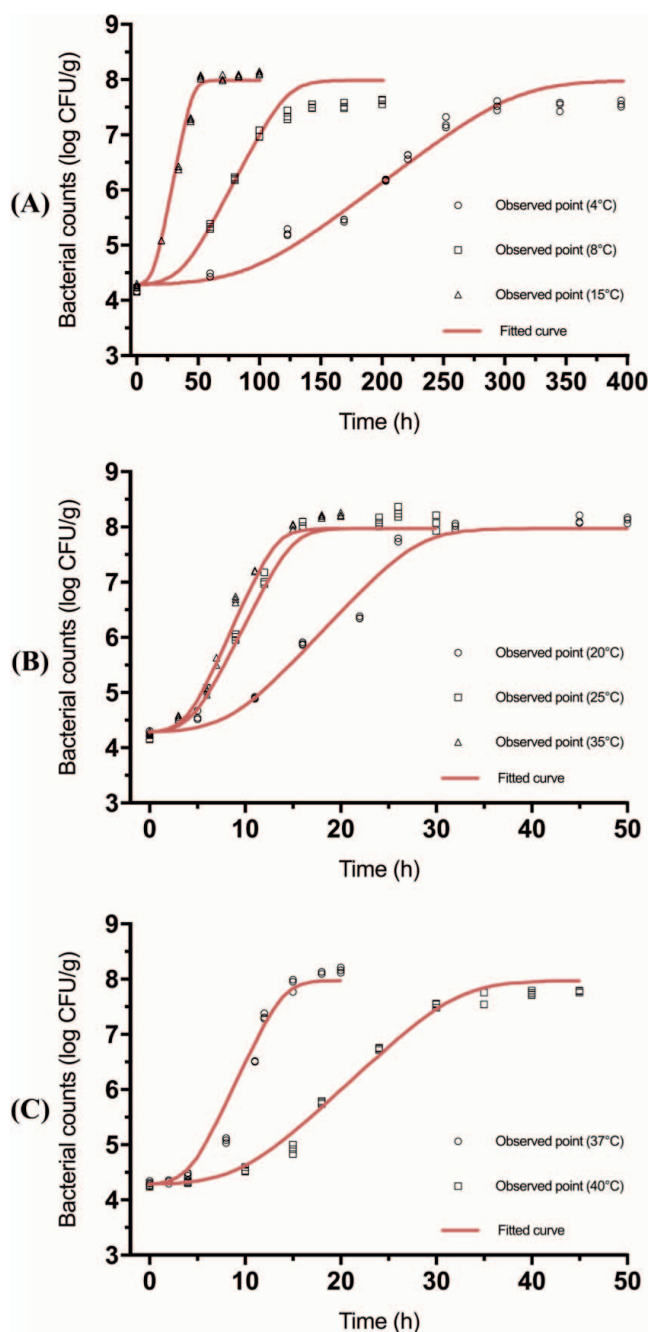


FIGURE 2. One-step fitting of the static model under eight static conditions. (A) 4, 8, and 15°C; (B) 20, 25, and 35°C; and (C) 37 and 40°C.

parameters of the dynamic model under FTC1. Thus, the design of the time-temperature profile or the enumeration point should be optimized, which might help improve the accuracy of the dynamic growth analysis.

The model evaluation results show that overall, the values of AICc for the regression evaluation of the one-step dynamic and static models were -310.12 and -298.85, respectively. The RMSE values for both regressions were within 0.25 log CFU/g, which indicated that both integrated models could well describe the observed data under the corresponding conditions.

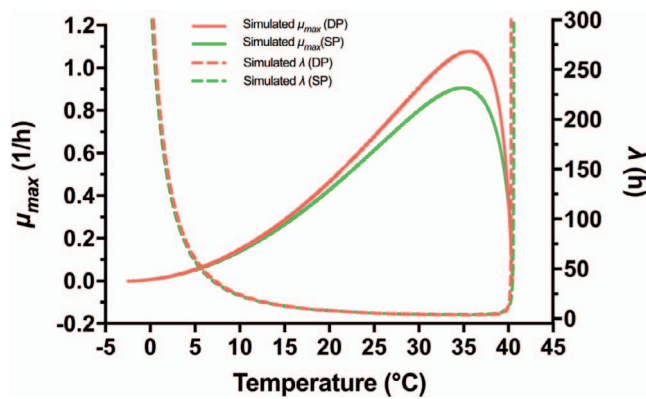


FIGURE 3. Effect of temperature on $\mu_{\max}$ and λ generated by the parameter derived from the one-step dynamic (DP) and static (SP) methods.

Temperature effects on the kinetic parameter. The effects of different temperatures on $\mu_{\max}$ could be simulated and compared by substituting the estimated parameters into the cardinal parameters model (Fig. 3, solid lines). Meanwhile, based on the estimated q_0 and $\mu_{\max}$, the lag effect at different temperatures could be determined by the function of λ in equation 3 and as shown in Figure 3 (dashed lines). The value of $\mu_{\max}$ simulated by SP was lower than that simulated by DP when temperature approximates are optimum, although it is not a significant difference considering the standard errors of the estimations. Both simulated λ curves were close in the entire growth temperature range. Meanwhile, when the beef samples were below 8°C or above 40°C, *L. monocytogenes* had a small $\mu_{\max}$ (lower than 0.1/h) and large λ (higher than 9.5 h). From the perspective of the food supply, the cold chain condition (ca. 4°C) might help inhibit *L. monocytogenes* growth in RTE braised beef during storage and transportation processes.

Simulations of the physiological state. Figure 4A and 4B shows the generated curves of $\alpha(t)$ at the FTC1 and static conditions of 4 and 35°C, respectively. Under FTC1,

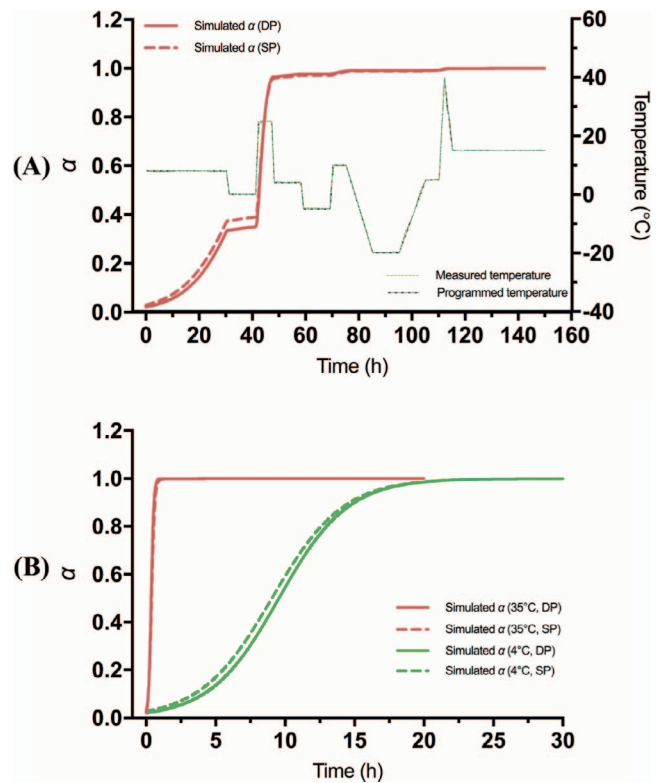


FIGURE 4. Simulated curves of α by the parameter derived from the one-step dynamic (DP) and static (SP) methods under (A) the dynamic conditions of FTC1 and (B) the static conditions of 4 and 35°C.

the change of α depended on the temperature fluctuations until the value approximated to 1. The dropped temperature could retard the accumulation of α . Under the static conditions, it was observed that the value of α developed to 1 without fluctuation, and the growth rate of α at 35°C was higher than at 4°C. According to the inference of the Baranyi model, the growth period of α at static temperatures should equal the corresponding lag time. The integrated dynamic Baranyi model might be used theoretically to determine the potential λ of an inoculum by introducing prior knowledge on its physiological state.

TABLE 1. Estimated parameters of the one-step dynamic regression (DP)

Parameter	Estimated value	Standard error	95% CI ^a		df ^b
			Lower	Upper	
μ_{opt} (per h)	1.078	0.179	0.720	1.437	65
T_{opt} (°C)	35.7	1.1	33.4	38.0	
T_{min} (°C)	-2.54	0.66	-3.87	-1.22	
T_{max} (°C)	40.5	3.2	34.0	46.9	
q_0 (-)	0.0209	0.0049	0.0133	0.0286	
Y_0 (ln CFU/g)	9.936 (4.31) ^c	0.049	9.839	10.034	
Y_{max} (ln CFU/g)	18.951 (8.22) ^c	0.034	18.884	19.019	
AICc	-310.12				
RMSE	0.10 (0.04) ^c				

^a CI, confidence interval.

^b df, degrees of freedom.

^c Converted to log CFU per gram.

TABLE 2. Estimated parameters of the one-step static regression (SP)

Parameter	Estimated value	Standard error	95% CI ^a		df ^b
			Lower	Upper	
μ_{opt} (per h)	0.913	0.027	0.860	0.966	206
T_{opt} (°C)	35.00	0.17	34.67	35.33	
T_{min} (°C)	-2.72	0.11	-2.94	-2.50	
T_{max} (°C)	40.699	0.056	40.588	40.810	
q_0 (-)	0.0314	0.0085	0.0148	0.0481	
Y_0 (ln CFU/g)	9.848 (4.28) ^c	0.084	9.682	10.013	
Y_{max} (ln CFU/g)	18.359 (7.97) ^c	0.062	18.238	18.481	
AICc	-298.85				
RMSE	0.48 (0.21) ^c				

^a CI, confidence interval.

^b df, degrees of freedom.

^c Converted to log CFU per gram.

However, a dispute surrounds the Baranyi model on the assumption that q could only reflect the accumulation of the critical substance in cells (3). In other words, when enough of the critical substance has accumulated (α approximates to 1), the lag time will not exist in the bacterial growth period, which might conflict with some observations (11). In addition, the variability of bacterial cells and temperatures might cause different lag behaviors at the single-cell level (2, 28, 40). Hence, further research is needed to validate the simulation of q and potential λ during the fluctuating temperature. Mechanisms remain to be explained for the physiological state and lag effect of the microbial inoculum, especially under dynamic conditions.

Model validation. The temperature will not remain static during the real food supply process (26). It is thus reasonable and necessary to validate microbial growth under dynamic conditions. In this study, DP and SP were introduced in equation 1 to validate the growth of *L. monocytogenes* in RTE braised beef at the additional conditions of FTC2, FTC3, and FTC4. The initial value of Y_0 for each profile was set to be the average *L. monocytogenes* concentration at time zero. Other parameters were set according to the corresponding estimation in Tables 1 and 2.

A total of 264 growth data were observed under FTC2, FTC3, and FTC4 and used together for validation. Similar to the observations under FTC1, the low initial temperature of FTC2 and FTC4 can prolong the lag period and restrain the growth of *L. monocytogenes* (Fig. 5A and 5C). The value of Y_0 in the growth under FTC4 was nearly 1 log less than under other experiment groups, but it had no obvious effects on the overall simulation. Although the endpoint temperature of FTC3 is set at 4°C, the MPD could reach 8.00 CFU/g (Fig. 5B). In contrast to the earlier observations under the isothermal conditions, however, no evidence illustrated that the refrigeration temperature has a significant influence on the MPD level under dynamic temperatures. This inconsistency is noteworthy, especially in real fluctuating cold chain conditions. Thus, from the perspective of food safety, it is not recommended to exclude the potential that microorganisms reach the highest concentration at low temperatures.

In all cases, the RMSE values of the prediction performed by DP and SP were 0.19 and 0.23 log CFU/g, respectively. Meanwhile, the bias factor values were 0.9893 and 1.0149 for the dynamic model developed by DP and SP, respectively. The parameter group of SP might slightly underestimate the bacterial concentration when the growth approaches the asymptotic phase (Fig. 5), which could potentially lead to a fail-dangerous prediction. But in practice, a high *L. monocytogenes* load is uncommon in RTE foods. Therefore, it is undeniable that the parameter groups derived from both one-step analysis methods could be conducted to accurately predict the dynamic growth of *L. monocytogenes* in RTE braised beef under FTC2, FTC3, and FTC4.

In conclusion, this work applied the one-step dynamic and static methods to determine the growth parameters of *L.*

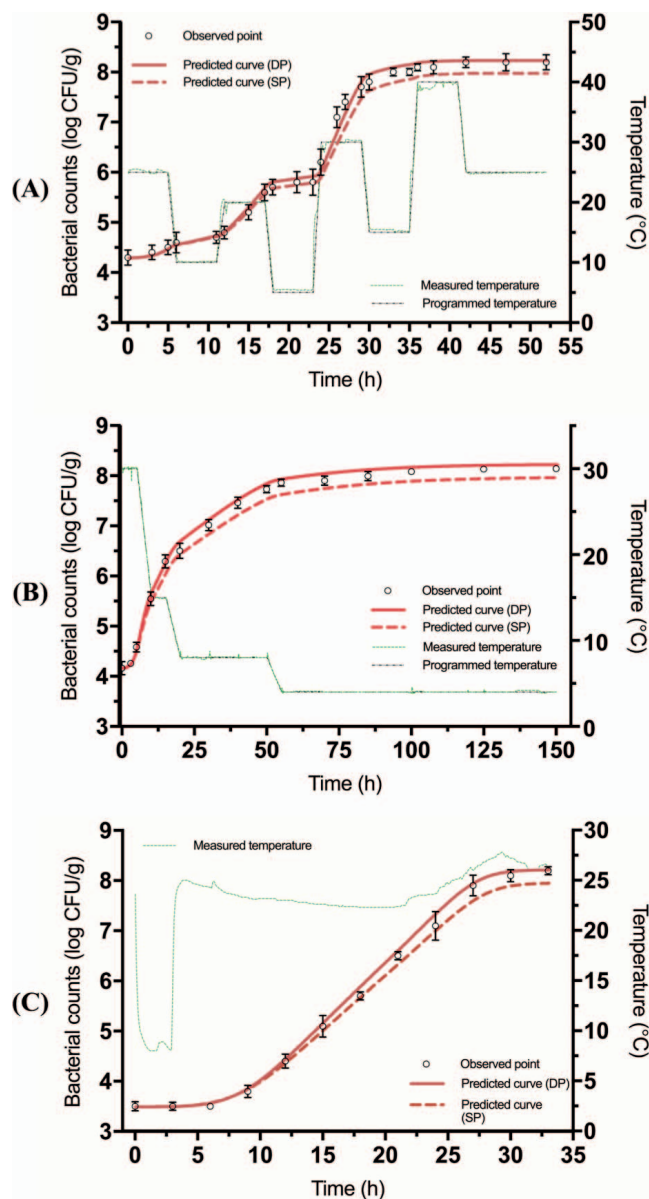


FIGURE 5. Validation of *L. monocytogenes* growth under the dynamic conditions of (A) FTC2, (B) FTC3, and (C) FTC4.

monocytogenes under dynamic and static temperature conditions, respectively. The parameters derived from both methods were acceptable for predicting the dynamic growth of *L. monocytogenes* in RTE braised beef. Comparing the model validation, the one-step dynamic analysis led to a slightly better prediction under varying temperatures than the one-step static analysis, especially near the MPD. In future studies, the design of the time-temperature profile and enumeration point for the dynamic analysis need to be optimized to obtain the estimated parameters with fewer standard errors. Meanwhile, interactions among different environmental conditions and microorganisms should be considered for better understanding the bacterial physiological states and MPD under various conditions, which might facilitate dynamic modeling for microorganisms in a complex ecosystem.

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